

SATQUERY-AI MISSION REPORT

GEOSPATIAL BI-TEMPORAL CHANGE DETECTION & EVIDENCE REPORT • SMART INDIA HACKATHON

Mission ID:	test_mission_report_analysis	Generated:	2026-09-08T20:49:11
Mode:	bi_temporal	Detector Threshold:	0.70 (LEVIR-CD Calibrated)
Checkpoint:	best_model.pt (SiameseUNet)	Confidence:	N/A — Uncalibrated

1. EXECUTIVE SUMMARY

SatQuery-AI completed bi-temporal change analysis for mission ID test_mission_report_analysis. The Siamese U-Net detector identified 9.09% changed pixels (71,495 of 786,432 total pixels) between the supplied acquisitions.

2. VISUAL EVIDENCE & SENSOR APERTURE

		Siamese U-Net Change Heatmap
T1: Before Acquisition [Image asset unavailable on disk]	T2: After Acquisition [Image asset unavailable on disk]	

3. QUANTITATIVE CHANGE DETECTION & SPATIAL ANALYTICS

Total Scene Pixels	786,432	Changed Pixels	71,495 (9.09%)
Unchanged Pixels	714,937 (90.91%)	Model Threshold	0.70
Coordinate System (CRS)	Not available	Spatial Resolution (GSD)	Not available
Physical Changed Area	Unavailable (No reliable GSD)	Total Physical Scene	Unavailable (No reliable GSD)

4. CONNECTED COMPONENT HOTSPOTS (TOP LOCALIZED CLUSTERS)

Cluster #	Pixel Area	Share of Change	Centroid (px: X, Y)	Bounding Box (px: x1, y1, x2, y2)
Cluster 1	1,200 px	N/A	(150.0, 200.0)	[100, 180, 200, 220]

5. REGION OF INTEREST (ROI) INVESTIGATION

ROI investigation was not performed.

6. AI VISION-LANGUAGE INTERPRETATION (CHANGE VQA)

Status: Executed via Domain-Adapted VLM	Confidence: N/A — Uncalibrated
AI Interpretation: Vegetation clearing observed along the southeastern corridor.	Evidence Grounding: • Bi-temporal contrast localized across coordinates.

7. EXECUTION TRACE & MODEL PROVENANCE

8. SENSITIVITY, UNCALIBRATED METRICS & LIMITATIONS

- Confidence is not calibrated across detector and vision-language models (strictly reported as N/A — Uncalibrated).
- Change detection operates on pixel-level contrast via Siamese U-Net without assigning semantic land-cover categories.
- Physical area unavailable — reliable spatial resolution (GSD) metadata was not provided in image headers.
- Geospatial coordinates unavailable — analysis spatial coordinates are reported strictly in pixel space.
- ROI investigation was not performed for this report compilation.